

Results, Conclusions, Future Work

CarSight AI

Smarter Used Car Pricing Through Damage Intelligence

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1 CONCLUSIONS

1.1 Changes in Development

To begin with, the initial hypothesis of the project was quite straightforward – to develop a price prediction model based on structured data. However, during the development process, it became apparent that there was no gap for the structured part of the data, and thus the project had to pivot towards the other gap – that of interpreting the textual information about the damages to the car.

This shift resulted in integrating natural language processing into the model and became the key element of the project. Another important decision was replacing the model of one predicted value with the prediction range. At first, I considered developing the model that would predict one value to be more realistic and feasible. However, in reality, the price could never be predicted precisely; thus, having a price range made more sense.

Finally, the chatbot was added quite late to the project, and it was not planned initially. However, including this feature helped make the project more complete and user-friendly.

1.2 Challenges

The biggest problem was handling the unstructured textual data. While it is relatively easy to deal with structured input, damage descriptions varied from user to user in terms of language used and sometimes even the terminology employed. In addition, such phrases as 'minor problem' or 'it runs fine' are quite difficult to quantify.

Data quality was another problem I had to tackle. Due to the nature of my research, I could only use publicly available information which inevitably contained some errors, such as mispriced cars or inconsistent descriptions.

Finally, time constraints were also an issue. It took considerable time to study the basics of NLP and implement it in a working solution. Creating both the backend and the front-end of the system took much longer than expected.

1.3 Lessons Learned

One lesson I learned was that practical problems have messy datasets. Working with incomplete, noisy, or imperfect data was just as crucial as modeling itself.

Another thing I learned was the significance of interpretability. It wasn't sufficient for our model to output a result; people needed an explanation behind the prediction. This is why we decided to include explanations in the form of breakdowns, as well as a chatbot.

The final lesson was that implementing an entire system from data processing through to the interface is fundamentally different from designing and creating the model alone.

2 FUTURE WORK

2.1 Model Improvements

If time allowed, one thing I would consider improving is the model architecture used. Currently, I'm relying on basic techniques such as TF-IDF and embeddings. Instead, I could apply a more sophisticated solution like transformers or BERTs for modeling language nuances.

In addition, I would make an effort to increase labeling of the damage severity attribute, currently made by humans following rules but potentially learnable by the model using labeled data.

2.2 Better Data & Real-World Testing

The main problem with this model is the lack of relevant and high-quality data. If I was able to pursue this project further, the first step I would take is acquiring more datasets, possibly from API marketplaces.

Another task that could be pursued is assessing the model on real listings from various marketplaces instead of synthetic data generated artificially for evaluation purposes.

2.3 Image + Text Integration

An important improvement would be using images alongside text.

Currently, the model relies solely on text features. However, in practical usage scenarios, visual information contained in images plays a vital role. Computer vision algorithms should be applied for detecting damage present in images and then combined with textual analysis results.